Air Monitor
Precision Airflow Measurement
An ONICON Brand

Accurate Airflow Measurement for Commercial HVAC Applications







PRECISE, RELIABLE, TRUSTED
SOLUTIONS

- BUILDING AIRFLOW MEASUREMENT AND CONTROL
- ROOM OR SPACE PRESSURIZATION
- OUTSIDE REFERENCE PRESSURE MEASUREMENT
- LABORATORY & HOOD EXHAUST
- OUTDOOR AIRFLOW MONITORING

We believe in our products, so should you.
1 YEAR NO FAULT WARRANTY
3 YEAR MANUFACTURER WARRANTY

FACTORY CONFIGURED

ON TIME DELIVERY

BEST CUSTOMER SERVICE

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HVAC APPLICATIONS

BUILDING AIRFLOW MEASUREMENT AND CONTROL - The distribution of conditioned air is the primary means of heating and cooling most commercial buildings today. Proper airflow control within a building is not only important for the health and comfort of the building's occupants, buy also for the health and long term performance and longevity of the building. Accurate airflow measurement allows the HVAC system to work effectively as designed, and efficiently as required to meet ever increasing energy conservation goals and high-performance building standards. Installations for this application include:

Ducted Airflow Measurement - Mechanical duct work is the most common means for distributing the conditioned air throughout most commercial buildings. The duct system provides the best opportunities for accurate airflow measurement due to its controlled dimensions, and the typical obstructions found within a duct system are generally well defined.

Fan Inlet/Discharge Measurement - Measuring airflow at the fan inlet presents challenges from a measurement technology stand point. Achieving accurate airflow measurement at the fan inlet, without affecting fan performance, is an important consideration when selecting the correct meter. The large variation in airflow velocity, as well as the multiple fan configurations that exist within an AHU, must be taken into consideration. Measuring airflow at the fan inlet can be advantageous from an accessibility stand point as well as monitoring the performance of fan walls at the source.

Outdoor Airflow Measurement - Controlling the amount of outside air entering a building is required to maintain pressurization, meet energy efficiency goals, confirm compliance with local building codes, and maintain the health of the building and its occupants. Accurate measurement of outside airflow is required for proper operation of today's high performance buildings. Outside air can pose a significant challenge regarding the metering technology to be selected. Some of the more common challenges associated with outdoor airflow measurement include: low airflow velocities over large operating ranges such as a split - min/max economizer system, blowing dust and debris, and moisture laden air at the point of measurement. Choosing a technology that is impervious to airborne contaminants, can measure low airflow rates, has high turn down capability and one that provides the desired BAS outputs will facilitate a successful installation and accurate outdoor airflow measurement.

		THERMAL DISPERSION AIRFLOW MEASUREMENT			DIFFERENTIAL PRESSURE		
					FIXED INLET	STATIC PRESSURE SENSING	
Applications	Installations	ELECTRA-flo/SD	ELECTRA-flo/S5 Probe Array	ELECTRA-flo/FI Fan Inlet Probe Array	OAM II Outdoor Airflow Measurement System	S.A.P. Static Air Probe	S.O.A.P. Static Outside Air Probe
Building Airflow Measurement and Control	Ducted Airflow	✶	✶				
	Fan Inlet			✶			
	Fan Discharge	✶	✶				
	Outdoor Airflow Monitoring		✶ (Recommended OAM II)		✶		
Building / Space Pressurization	Fan Tracking			✶			
	Outdoor Pressure Reference					✶	✶
	Room / Space Pressurization					✶	
Indoor Air Quality	Outdoor Airflow Monitoring		✶ (Recommended OAM II)		✶		
	Laboratory Hood Exhaust Duct						





BUILDING/SPACE PRESSURIZATION - Proper pressurization of buildings and indoor spaces is a crucial component required for the management of indoor air quality, maximizing energy efficiency and maintaining occupant health and comfort. Lack of control with regard to pressurization can lead to a host of problems including the infiltration of moisture, cold winter or hot summer drafts, and doors that are difficult to open or slam shut. Some airflow measurement strategies for this application include:

Fan Tracking - Fan tracking is the accurate measurement of the entire airflow system including supply and return air and outside and relief air. Airflow measurement is a much more effective and accurate means of maintaining building pressurization when compared to static pressure measurements made throughout an entire building.

Outside Reference Pressure Measurement - Indoor pressures must be measured relative to a reference pressure, usually the outside air pressure.

Room or Space Pressurization - This is essential for managing indoor air quality, energy savings and occupant comfort. Prevents unintended pressure levels which can lead to complications for the building systems, the building envelope, and problems between adjacent spaces such as laboratories and hospital rooms.

INDOOR AIR QUALITY - Creating and maintaining proper indoor air quality in today's built environment can be a challenging task. The effect that poor indoor air quality has on building occupants can range from loss of efficiency and performance to specific acute health issues. Airborne contaminants found within buildings come from a number of sources including people, processes and the materials used in building construction. Maintaining indoor air quality requires accurate airflow measurement. A few of the typical strategies implemented for this application include:

Outdoor Airflow Monitoring - Bringing in the right amount of outside air is crucial to maintain proper building operation, meet energy conservation goals, and maintain the IAQ demanded in today's built environment. Providing adequate dilution air to the occupied space within a building is the best way to control the level of contamination within the space. Accurately and continuously monitoring the outside air flowing into a building will allow the BAS to control the building as designed.

Laboratory & Hood Exhaust - Provides essential information to the lab system for maintaining occupant comfort and safety, space pressurization relative to the rest of the building and/or other spaces, and confirms fume hood operation.

DIFFERENTIAL PRESSURE PITOT TUBE VELOCITY PRESSURE AIRFLOW MEASUREMENT				TRANSMITTERS AND MONITORS			
VOLU-probe Pitot Traverse Probe	FAN-E Pitot Traverse Station	VOLU-probe/FI Pitot Fan Inlet Traverse Probe	LO-flo Pitot Traverse Station	VELTRON DPT 2500-plus	VELTRON III	Installations	Applications
✶	✶		✶	✶	✶	Ducted Airflow	Building Airflow Measurement and Control
		✶		✶	✶	Fan Inlet	
✶	✶			✶	✶	Fan Discharge	
				✶	✶	Outdoor Airflow Monitoring	
		✶		✶	✶	Fan Tracking	Building / Space Pressurization
				✶	Coming Soon	Outdoor Pressure Reference	
				✶	Coming Soon	Room / Space Pressurization	
						Outdoor Airflow Monitoring	Indoor Air Quality
✶(SS)	✶(SS)		✶(SS)	✶	✶	Laboratory Hood Exhaust Duct	







ELECTRA-flo 5 Series Thermal Airflow Measurement System

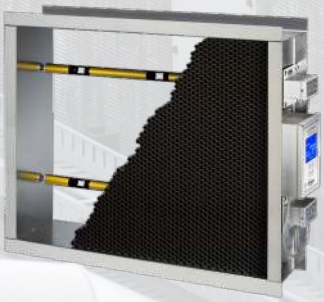
- Rugged probes with aerodynamic sensor apertures - Requires less straight run
- Up to 32 individual sensing points per transmitter - More sensing points means better accuracy
- Daisy chain multiple probes per transmitter - Reduces cabling and conduit
- NIST* traceable calibration - Guarantees accuracy within ±2% of actual flow
- ELECTRA-flo G5 Transmitter included - Local display with programming interface, BACnet® MS/TP, MODBUS® and analog outputs. Optional true dual channel version with duct size, dimension data and sensing point allocation set independently for each channel
- Robust sensor design provides protection against environmental conditions and allows damage-free cleaning



ELECTRA-flo/SD Thermal Airflow & Temperature Measurement System

Designed to measure airflow and temperature in small duct variable air volume applications.

- Measure airflow and temperature in 4”-16” diameter ducts
- Analog outputs for flow and temperature for improved control and efficiency in multi-zone VAV systems
- Optional BACnet or MODBUS RS485



ELECTRA-flo/CM Thermal Airflow Measurement Station

All features of ELECTRA-flo 5 Series Thermal Airflow Measurement System plus:

- ELECTRA-flo/S5 thermal probes mounted in rigid, welded, galvanized casing - Simplifies installation
- Honeycomb cell air straightener - Reduces straight run requirements



ELECTRA-flo/FI Thermal Dispersion Fan Inlet Probe Array

- Dual point thermal dispersion sensor probes mounted in sensor housing - Simplifies installation
- Installs directly in fan inlet with virtually no pressure drop
- ELECTRA-flo G5 Transmitter included

*National Institute of Standards and Technology





Robust construction and reliable readings that are unaffected by wind direction, airborne moisture, and dirt make the OAM II the ideal solution for outdoor airflow applications!



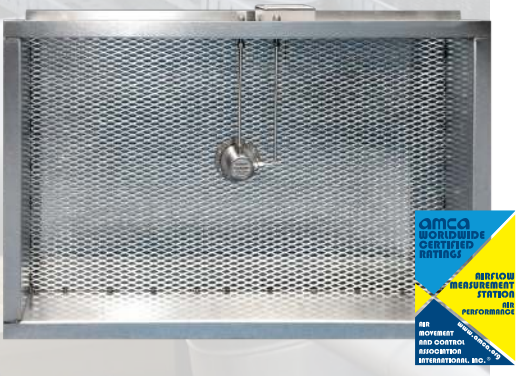
OAM II Outdoor Airflow Measurement System

- Robust and reliable construction - Readings are unaffected by wind direction, airborne moisture and dirt
- Factory calibration - Guarantees accuracy within ±5% of reading
- Measures inlet velocities as low as 100 FPM
- New and retrofit installations onto most single and dual inlet package air handlers
- Local display of data and direct analog interface with BAS for data logging and/or control of outside air dampers
- ASHRAE 62 and 189.1 compliant



OAM II Outdoor Airflow Measurement Uni-Sensor

- Unaffected by airborne dust and debris and condensing moisture
- Reliable airflow measurement without the need for long straight runs
- Largely immune to effects of shifting wind loads and wind gusts
- The Uni-Sensor is constructed of 316 Stainless Steel and is resistant to corrosion caused by salt and most other airborne corrosives
- Uni-Sensors are ideal for retrofit installations because they are easily mounted to a variety of existing inlet types
- Uni-Sensors combine an outside reference (high pressure) sensor and an inlet airflow (low pressure) sensor into one assembly



OAM II Outdoor Airflow Measurement Station

- Stainless steel sensors factory mounted directly onto rugged casing - Simplifies installation
- Perforated metal mesh provides known fixed inlet flow coefficient - Removes the need for field characterization
- Factory calibrated for selected applications - Guarantees accuracy
- AMCA* certified - Within ±5% of actual airflow across specified flow range

*Air Movement and Control Association

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VOLU-probe Pitot Airflow Measurement Traverse Probe

- Multiple Pitot total and static pressure sensing points - Improved accuracy
- Senses average total and static pressure traverses of an air stream
- AMCA* certified - Within ±2% certified accuracy

VOLU-probe/SS Stainless Steel Airflow Measurement Traverse Probe

- Ideal for clean or harsh and particulate laden applications
- Temperature range is -2°F to 900°F
- Accurate within ±2-3% of actual flow





VOLU-probe/VS Pitot Airflow Measurement Traverse Station

- One or more VOLU-probes factory mounted in a rigid, galvanized casing - Simplifies installation
- Senses average total and static pressure traverses of an air stream
- AMCA* certified - Within ±2% certified accuracy

VOLU-probe/FI Pitot Fan Inlet Airflow Traverse Probe

- Pair of offset traverse probes mount directly to fan inlet
- Accurately measures inlet velocity pressure and air volume
- Aluminum or stainless steel





FAN-E Pitot Airflow Measurement Traverse Station

- Multiple Pitot total and static pressure sensing points
- Improved accuracy
- Traverse station includes air straightening honeycomb cell
- Reduces straight run requirements
- AMCA* certified - Within ±2% certified accuracy

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VELTRON DPT 2500-plus Transmitter

- Ultra-low differential pressure and flow transmitter
- Accurate within ±0.25% of natural span - Ranges from 0.05 to 25.0 in wc
- Ideal for demanding HVAC applications
- Microprocessor based configuration and calibration

VELTRON III Transmitter

- ±3% of the reading when combined with AMC sensors
- Stacked transducers for higher accuracy over a wider turndown
- Dual channel configuration to measure two separate applications with one transmitter
- Field characterization for better performance in limited straight run
- Air density and relative humidity compensation
- BACnet and MODBUS Communication





Aluminum LO-flo Pitot Airflow Measurement Traverse Station

- Measures airflow in small round duct work between 4”- 8”
- Measures volume rates between 35 - 1700 CFM
- Accurate within ±2% of actual airflow

S.O.A.P. - Static Outside Air Probe

- Accurate and instantaneous sensing of outside static air pressure levels
- Unaffected by wind direction or gusts





S.A.P. - Static Air Probe

- Steady, non-pulsating output of room, space or plenum pressure measurements
- Aluminum or stainless steel construction

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SINCE 1967



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AMCA Certified Airflow Stations guarantee compliance with ASHRAE 189.1 High Performance Building Standards.



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Visit our website to find your local expert representative.

We believe in our products, so should you.

1 YEAR NO FAULT WARRANTY

3 YEAR MANUFACTURER WARRANTY

DOC-0002855 Rev K September 2024